Monash FIT3143 Parallel Computing | Applied #2: GPU/NPU and Exponential Growth

GPU Image Rotation and the Ethics of Scaling HPC for AI

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Task 1: GPU design

Task 2: Ethics of HPC

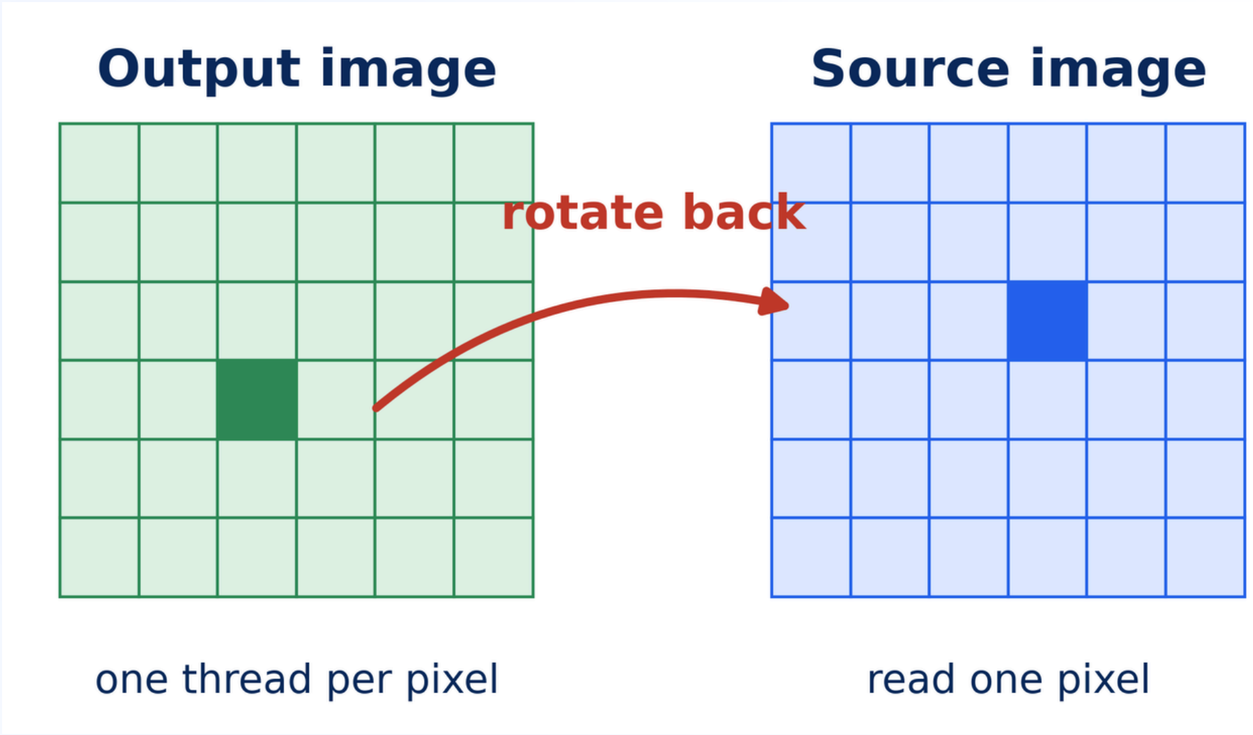
Limits and Q&A

Presenter: Taabish

Task 1: 33 Million Independent Pixels

- 8K = 7680 x 4320 = 33,177,600 output pixels
- Each pixel is independent: a data-parallel workload
- Inverse mapping: one thread per output pixel
- One writer per pixel, no holes

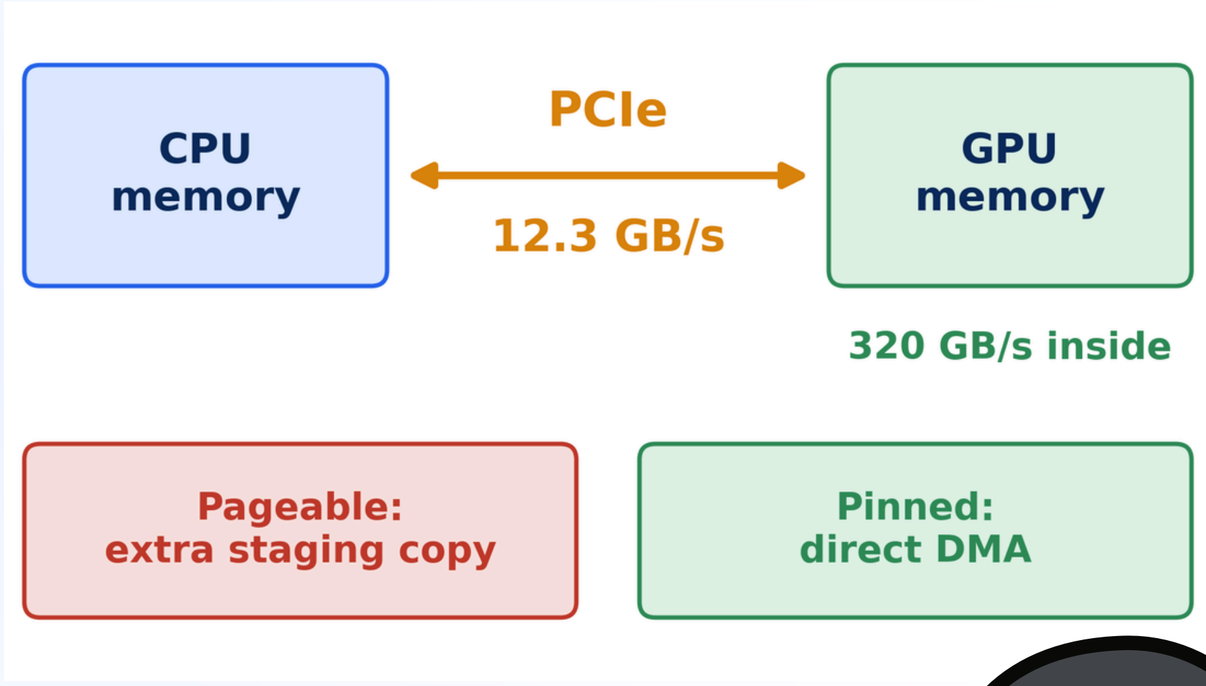




Presenter: Taabish

1a. Host to Device and Back

- Pageable: extra staging copy (4.6 GB/s)
- Pinned: DMA straight over PCIe
- Pinned H2D measured: 12.3 GB/s; T4 GDDR6: 320 GB/s (NVIDIA, n.d.-d)
- Zero-copy reads host RAM over PCIe on every access (NVIDIA, 2026a, 2026b)



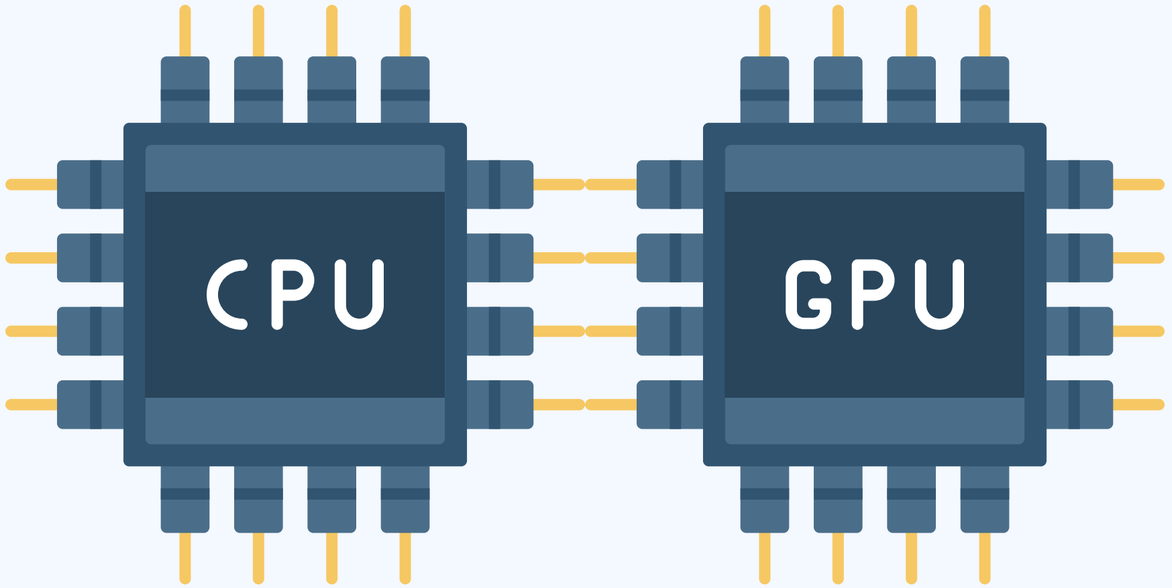


Sources: NVIDIA (n.d.-d, 2026a, 2026b). Measured on Tesla T4.

Presenter: Taabish

1b. CUDA Model: host, device, <<<grid, block>>>

- Host (CPU): allocate, copy, launch async, sync, check errors
- Device (GPU): SIMT threads in warps of 32
- Block 16 x 16 = 256 threads = 8 warps
- Grid 480 x 270 = 129,600 blocks for 8K



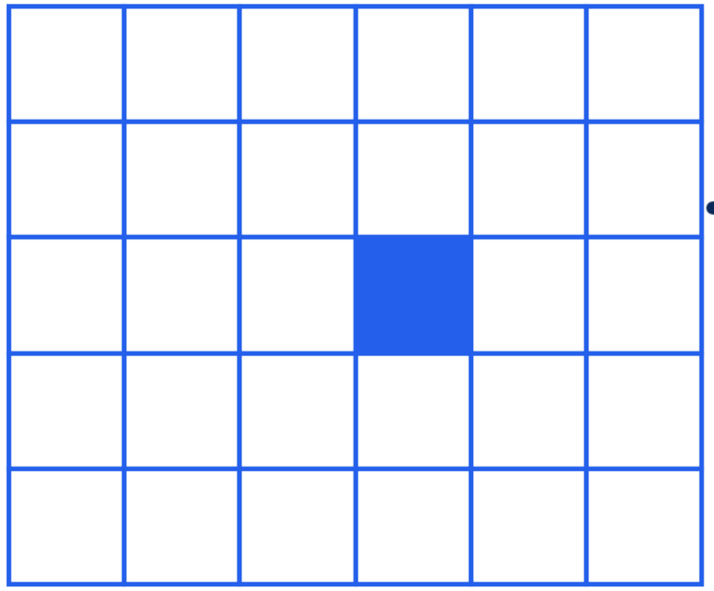


Sources: NVIDIA (2026b).

Presenter: Taabish

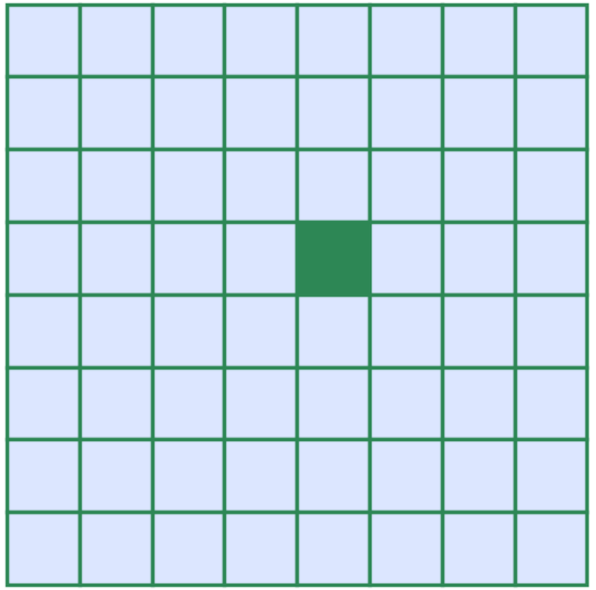
1b. CUDA model: grid, block, thread

Grid



480 x 270 blocks

Block



16 x 16 threads

= 8 warps of 32

1 thread
= 1 pixel



Sources: NVIDIA (2026b).

1b. Kernel and launch (from rotate_cuda.cu)

Presenter: Taabish

Kernel rotate_nn_2d: rotate_cuda.cu, lines 145 to 158

```
145  // v2: 2D grid of 2D blocks. blockIdx picks the output tile, threadIdx the
146  // pixel inside it. No divide or modulo like v1, which is why it is
147  // faster. Our sweep shows shape barely matters for nearest (128x1 ~ 16x16).
148  // y0..y1 lets the streams version run one band of rows at a time.
149  __global__ void rotate_nn_2d(const u8 *__restrict__ src,
150                               u8 *__restrict__ dst, int w, int h, Rot r,
151                               int y0, int y1)
152  {
153      int x = blockIdx.x * blockDim.x + threadIdx.x;
154      int y = y0 + blockIdx.y * blockDim.y + threadIdx.y;
155      if (x >= w || y >= y1)
156          return; // grid is rounded up to whole blocks
157      pixel_nearest(src, dst, w, h, r, x, y);
158  }
```

Grid size and launch: rotate_cuda.cu, lines 360 to 383

```
359  // Rounds the grid up so every pixel gets a thread.
360  static dim3 grid_for(dim3 block, int w, int rows)
361  {
362      // Round up so the ragged right and bottom edges still get threads.
363      return dim3((w + block.x - 1) / block.x, (rows + block.y - 1) / block.y);
364  }
365
366  // Launches the kernel for variant k on a band of rows in stream st.
367  static void launch(const Gpu &g, Kind k, dim3 block, cudaStream_t st,
368                     int y0, int y1)
369  {
370      dim3 grid = grid_for(block, g.w, y1 - y0);
371      switch (k) {
372      case K_NN_1D: {
373          int threads = block.x * block.y;
374          size_t n = (size_t)g.w * g.h;
375          unsigned blocks = (unsigned)((n + threads - 1) / threads);
376          rotate_nn_1d<<<blocks, threads, 0, st>>>(g.d_src, g.d_dst, g.w, g.h,
377                                                   g.r);
378          break;
379      }
380      case K_NN_2D:
381          rotate_nn_2d<<<grid, block, 0, st>>>(g.d_src, g.d_dst, g.w, g.h,
382                                               g.r, y0, y1);
383          break;
```

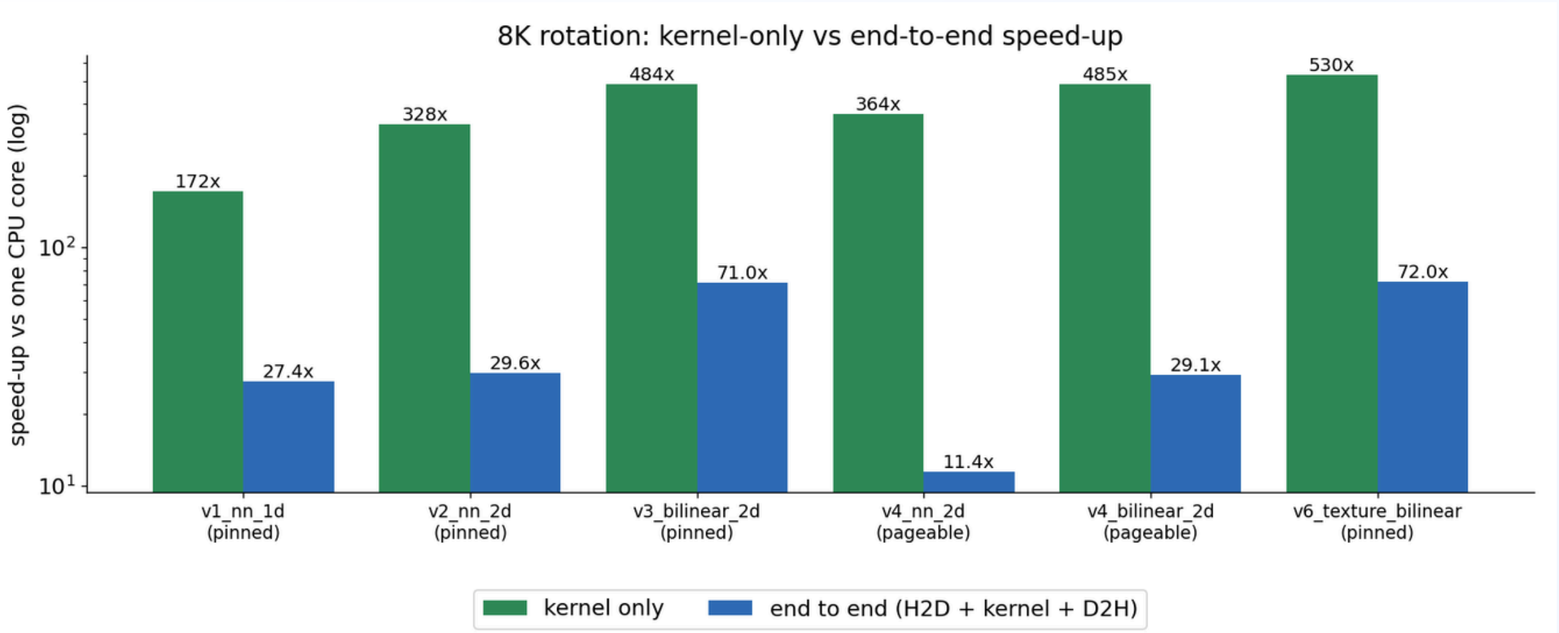
- Each thread owns one output pixel and reads its source with R transpose, so there are no holes and no write races
- Guard drops extra threads; grid is rounded up
- Launch is async, so we check the error right after
- Screenshots straight from rotate_cuda.cu in our submission



Sources: rotate_cuda.cu in our submission zip.

Presenter: Taabish

1b. Each Feature, Measured



328x
kernel only, vs 1 CPU core

29.6x
end to end

91%
of the time is copies

- 2D grid beat 1D by 1.1 to 1.9x (likely a 64-bit divide, not profiled)
- Pinned memory: copies 2.7x faster
- Streams: 1.81x on a batch of 8



Tesla T4, 8K at 30 degrees, v2 nearest, mean of 10 runs. CPU baseline: one core.

Presenter: Taabish

1b. Six versions, and what limits us

Why the speed-up stops

- Copies cap us at 32.6x (509.3 / (8.06 + 7.58)). We reached 29.6x, 91% of it
- H100 bandwidth is about 10x the T4, but PCIe copies do not speed up
- Against both Colab cores (OpenMP, 470 ms): still 27x
- Streams overlap copies and kernels

Streams code

Streams loop with cudaMemcpyAsync: rotate_cuda.cu, lines 803 to 815

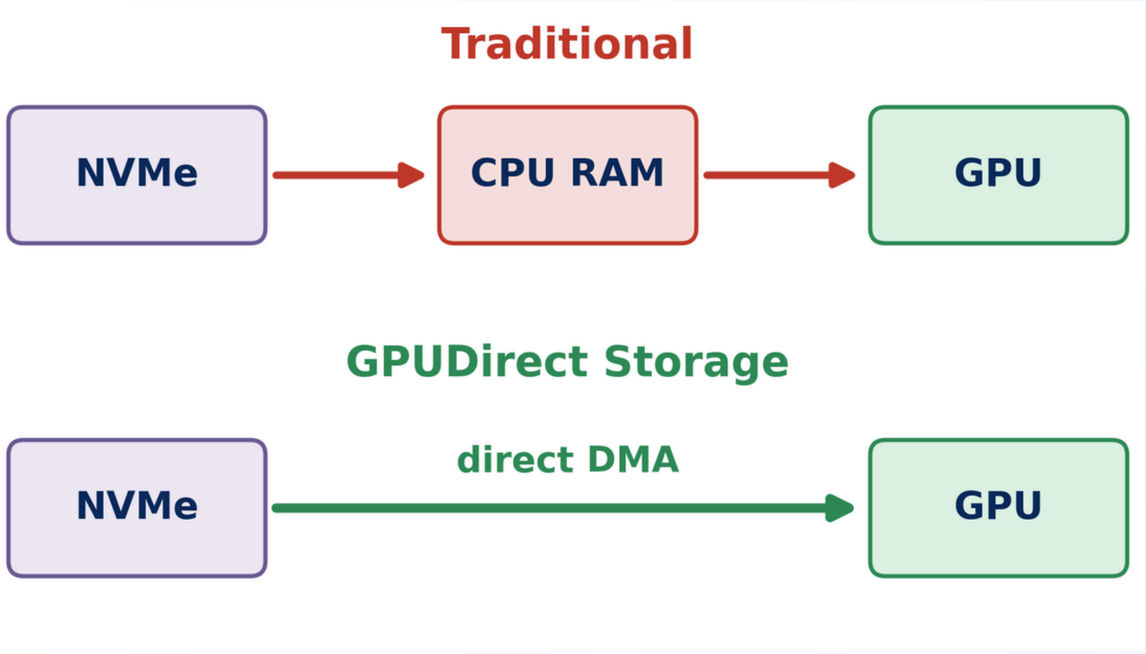
```
803         // Streamed batch: image i goes to stream i % nstreams, so one
804         // image's H2D can run beside another's kernel and a third's D2H.
805         CUDA_CHECK(cudaEventRecord(g.ev[2], 0));
806         for (int i = 0; i < batch; i++) {
807             int k = i % nstreams;
808             gs.d_src = d_in[k];
809             gs.d_dst = d_out[k];
810             CUDA_CHECK(cudaMemcpyAsync(gs.d_src, b_in + i * n, n,
811                                        cudaMemcpyHostToDevice, st[k]));
812             launch(gs, K_BL_2D, g.block, st[k], 0, g.h);
813             CUDA_CHECK(cudaMemcpyAsync(b_out + i * n, gs.d_dst, n,
814                                        cudaMemcpyDeviceToHost, st[k]));
815         }
```

v2 (nearest, 16 x 16), 8K at 30 degrees, Tesla T4, mean of 10 runs.



1c. GPUDirect Storage: helps the bulk pipeline, not one photo

Presenter: Taabish



Helps

- Bulk raw frames, or JPEGs decoded on the GPU (nvJPEG)
- A raw 8K frame is 99.5 MB, so reads are costly
- When I/O is the bottleneck (Thompson & Newburn, 2019)



Does not help

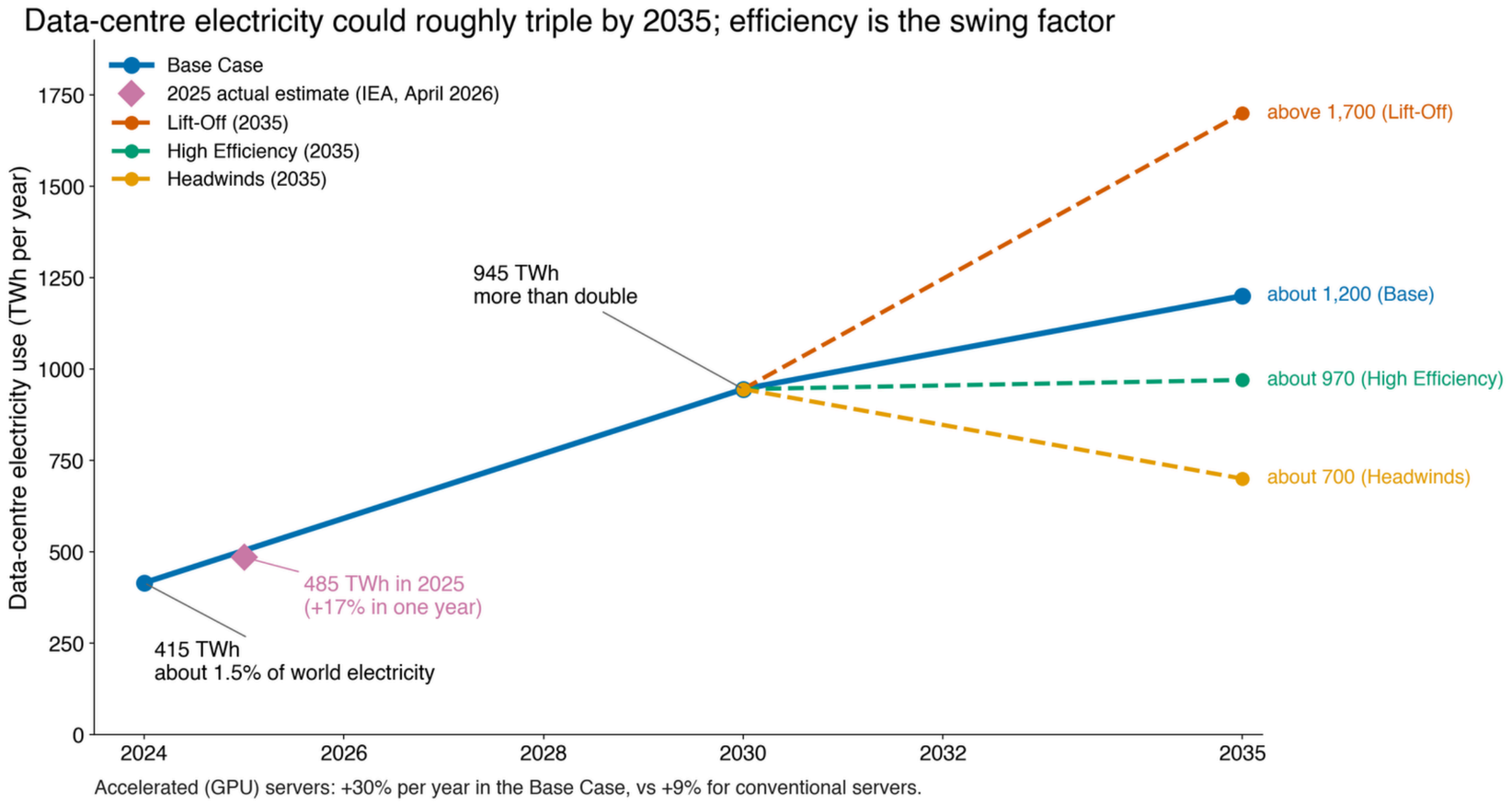
- One photo
- JPEGs decoded on the CPU
- Images already in RAM
- Needs nvidia-fs and O_DIRECT: we could not test it on Colab (NVIDIA, n.d.-a)

Not tested: Colab has no nvidia-fs module

Sources: Thompson & Newburn (2019); NVIDIA (n.d.-a).

Presenter: Erwyna

2a. Energy: data centre demand keeps climbing



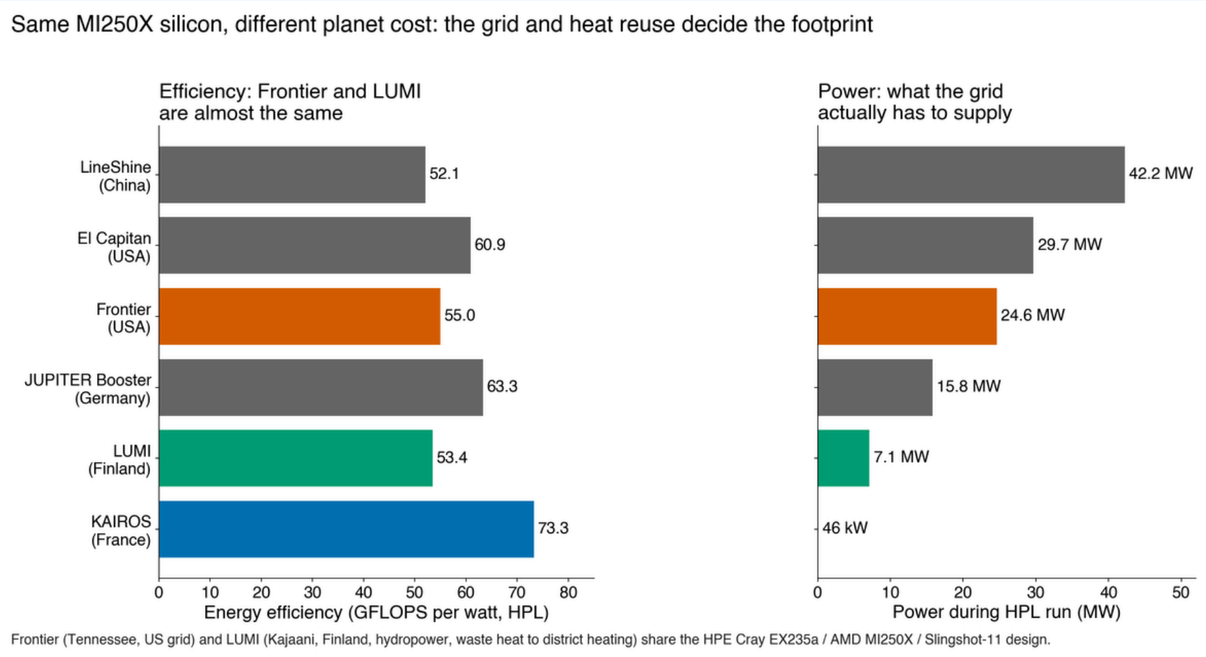
- 415 TWh in 2024
- 945 TWh by 2030, about 1,200 TWh by 2035
- GPU servers drive the growth (IEA, 2025)
- So we report energy per job



Sources: International Energy Agency (IEA, 2025).

2a. Same chip, different grid

Presenter: Erwyna



- Frontier: 55.0 GFLOPS/W
- LUMI: 53.4 GFLOPS/W
- LUMI: 100% hydropower
- Waste heat warms a town



Carbon per FLOP depends on the grid, not only the chip

Sources: TOP500 (2026); LUMI consortium (2023).

Presenter: Erwyna

2a. Malaysia lens and a myth check

Johor, Malaysia

- 79% fossil grid (Ember, 2026)
- 10 MW to 1.3 GW in 3 years (Loo, 2025)
- Up to 31% of Peninsular electricity by 2035 (Bernama, 2026)
- No new water-hungry Tier 1 and 2 sites (Shadiqe, 2025)

Kajaani, Finland (LUMI)

- 100% hydropower
- Cold climate cooling
- Waste heat warms the town (LUMI consortium, 2023)



Myth: "AI out-emits aviation"

It is a 2030 projection, not true today

Sources: Ember (2026); Loo (2025); Bernama (2026); LUMI consortium (2023); Jamison et al. (2025); Giles (2025); IEA (2025).

Presenter: Erwyna

2b. Data Governance, Security and Fairness

Governance

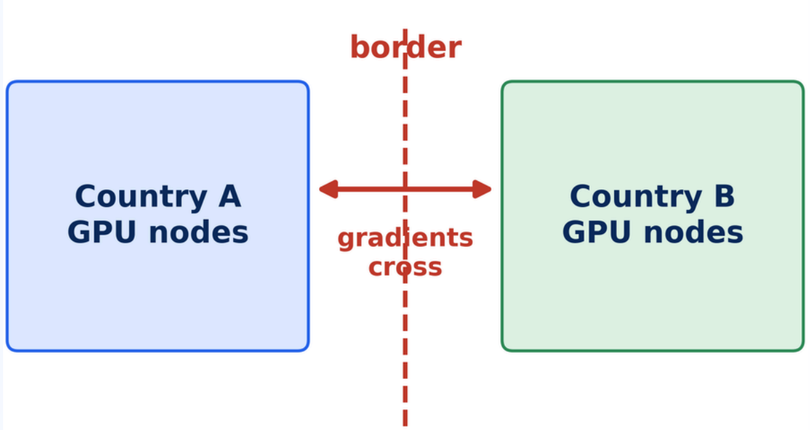
- 75% of AI supercomputer power is in the US (Pilz et al., 2025)
- Gradients can leak data (Zhu et al., 2019), so residency is a scheduling rule

Security: dual-use

- Same chips, hospital or military model: dual-use
- US limits exports; Malaysia now requires permits (BIS, 2022; MITI, 2025)

Fairness

- Malay is 0.086% of web text
- Tamil takes about 10 tokens per word (Common Crawl, 2026; Ahuja et al., 2023)

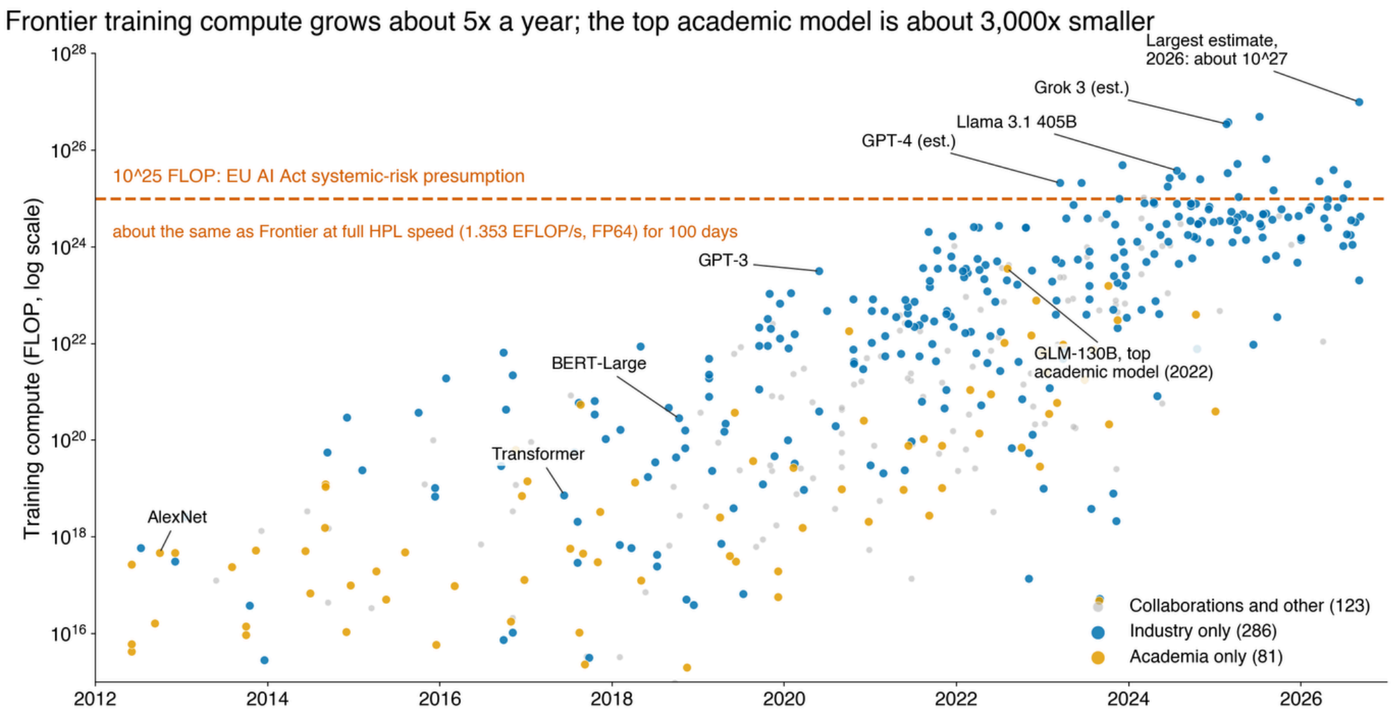




Sources: Pilz et al. (2025); Zhu et al. (2019); BIS (2022); MITI (2025); Common Crawl (2026); Ahuja et al. (2023).

Presenter: Erwyna

2c. The Compute Divide



Only companies can audit frontier models
NCI Gadi: 776 GPUs, demand about 3x allocation (NCI, n.d.-a, n.d.-b)

>90%

of notable models: industry

5x / year

frontier compute; top academic model about 3,000x lower

1 to 8 GPUs

for most academics



Sources: Stanford HAI (2026); Epoch AI (2026); Khandelwal et al. (2025); NCI (n.d.-a, n.d.-b).

2d. Our SCALE checklist

Presenter: Erwyna

Each step is a scheduler setting or a number

S Size right: stop adding GPUs when parallel efficiency drops

C Carbon-aware: up to 19% less CO2 for jobs that can wait

A Account every joule: energy per job, PUE reported

L Lawful, local data: keep data in its country

E Equitable access: fair-share queues

UNESCO, OECD, AIGE: values. EU: 10^25 FLOP, PUE above 500 kW



Sources: UNESCO (2021); OECD (2024); MOSTI (2024); European Parliament & Council (2024); Wiesner et al. (2021).

Limits, Future Work and Takeaways

Task 1

- Limits: Colab has only 2 vCPUs, shared VM, no GDS test
- Next: nvJPEG + GDS + streams on a GDS server
- Takeaway: memory bound, so PCIe sets the ceiling



Task 2

- Limits: model energy figures are estimates (Epoch AI, 2026); carbon-aware gains only 1 to 2% at peak hours (Radovanovic et al., 2023)
- Ours, measured: 0.71 J per 8K image, GPU board only (41 W for 17.2 ms)
- Takeaway: a FLOP's cost depends on where and for whom it runs

Sources: Epoch AI (2026); Radovanovic et al. (2023).

Presenter: Erwyna

Q&A

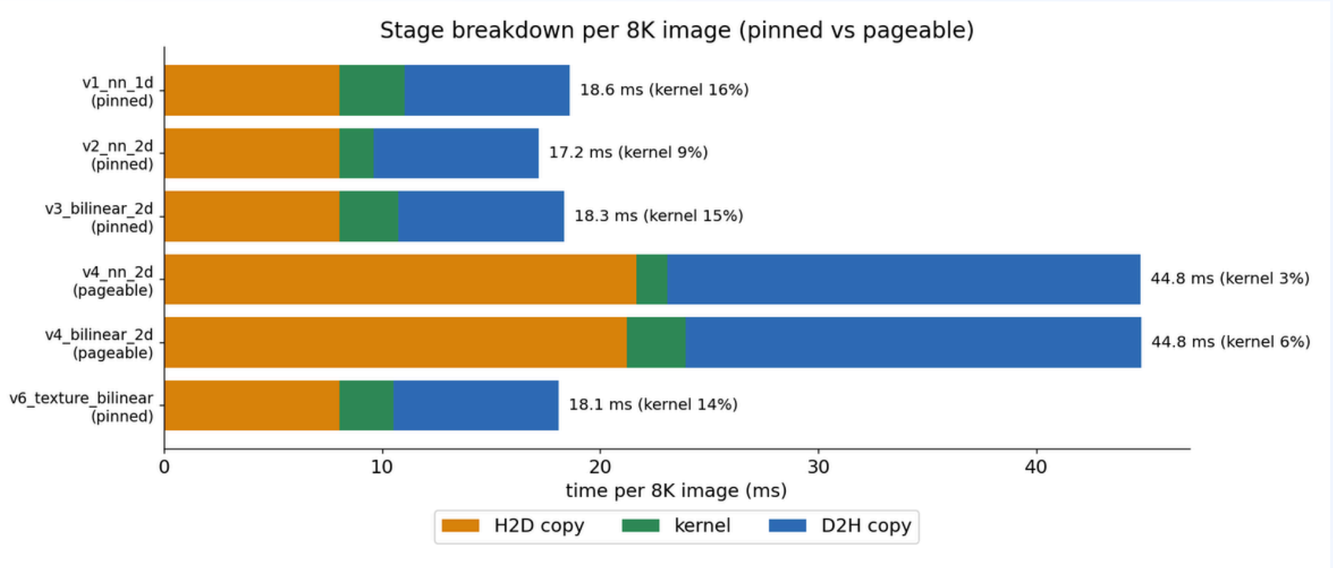
Questions?

Erwyna Soo Wen Xin and Taabish Farooq Bhat



A1. Where the time goes (8K, 30 degrees)

APPENDIX: extra material for Q&A, not part of the 7-minute talk



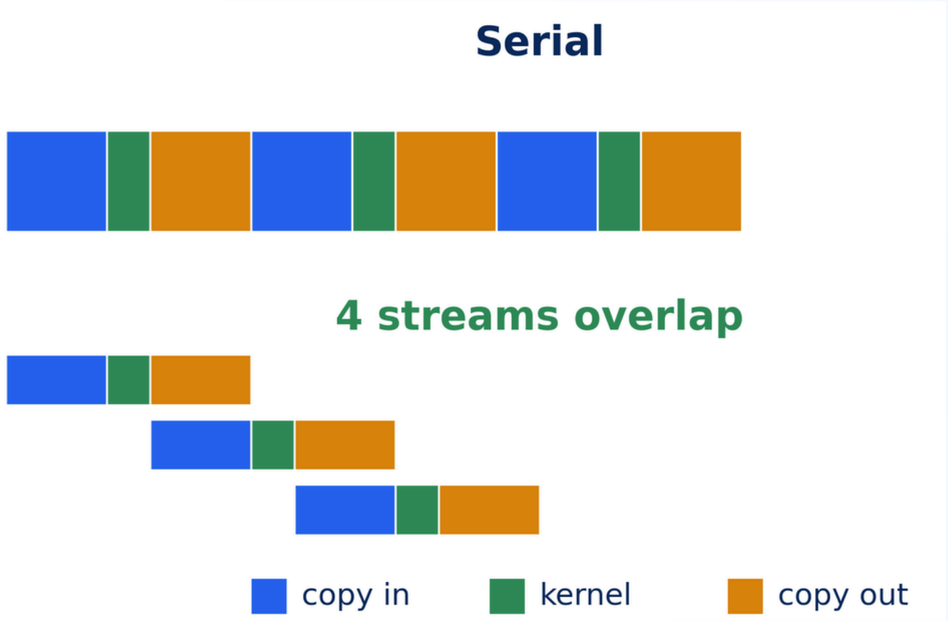
- H2D 8.06 ms
- Kernel 1.55 ms
- D2H 7.58 ms
- Total 17.19 ms
- CPU, one core: 509.3 ms
- Mismatch: 0 pixels



v2 nearest, pinned, Tesla T4, mean of 10 runs.

A2. Pinned vs pageable, and streams

APPENDIX: extra material for Q&A, not part of the 7-minute talk



Streams loop with cudaMemcpyAsync: rotate_cuda.cu, lines 803 to 815

```
803         // Streamed batch: image i goes to stream i % nstreams, so one
804         // image's H2D can run beside another's kernel and a third's D2H.
805         CUDA_CHECK(cudaEventRecord(g.ev[2], 0));
806         for (int i = 0; i < batch; i++) {
807             int k = i % nstreams;
808             gs.d_src = d_in[k];
809             gs.d_dst = d_out[k];
810             CUDA_CHECK(cudaMemcpyAsync(gs.d_src, b_in + i * n, n,
811                                        cudaMemcpyHostToDevice, st[k]));
812             launch(gs, K_BL_2D, g.block, st[k], 0, g.h);
813             CUDA_CHECK(cudaMemcpyAsync(b_out + i * n, gs.d_dst, n,
814                                        cudaMemcpyDeviceToHost, st[k]));
815         }
```

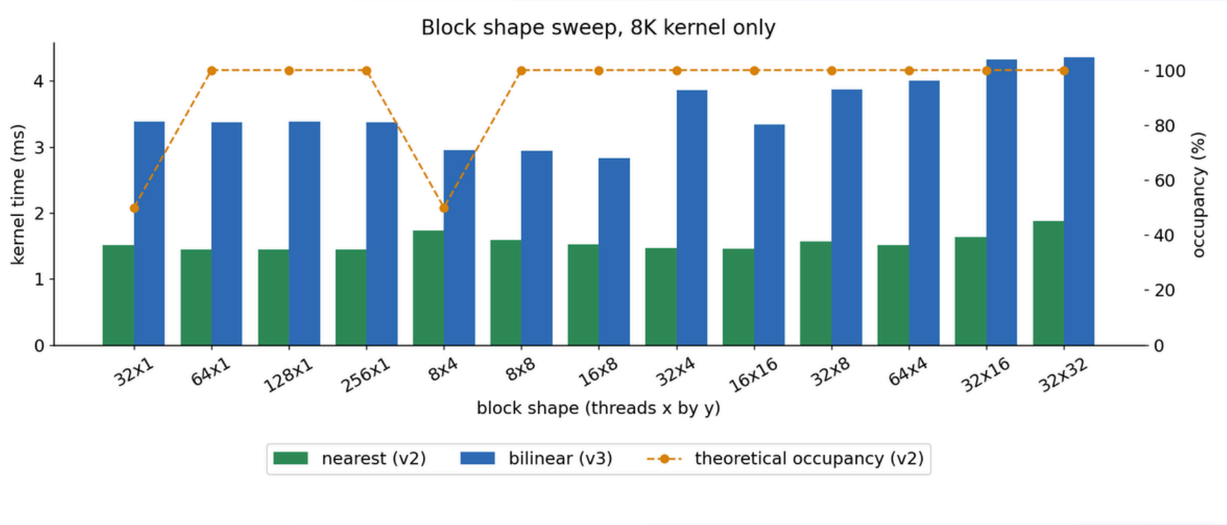
- Pageable 4.6 GB/s vs pinned 12.3 GB/s
- Streams: 1.8x on 8 images, 1.36x on one image

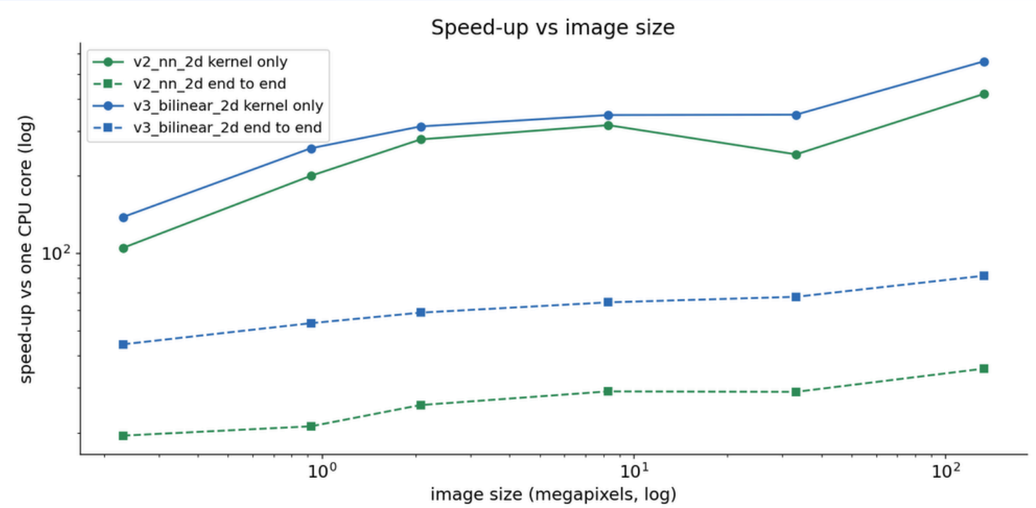


Image i goes to stream i % 4. Async copies need pinned memory. Source: rotate_cuda.cu in our submission zip.

A3. Block sweep and image size sweep

APPENDIX: extra material for Q&A, not part of the 7-minute talk





- Nearest: block shape barely matters
- Bilinear: 16 x 8 is fastest (2.83 ms)
- Bigger images: bigger speed-up



Run to run, the v2 kernel varied from 1.45 to 2.11 ms; copies stayed at 8.06 ms.

A4. How we timed it

APPENDIX: extra material for Q&A, not part of the 7-minute talk

CUDA events around H2D, kernel and D2H: rotate_cuda.cu, lines 440 to 450

```
440        CUDA_CHECK(cudaEventRecord(g.ev[0]));
441        CUDA_CHECK(cudaMemcpy(g.d_src, h_src, g.bytes,
442                              cudaMemcpyHostToDevice));
443        CUDA_CHECK(cudaEventRecord(g.ev[1]));
444        launch(g, k, g.block, 0, 0, g.h);
445        CUDA_CHECK(cudaEventRecord(g.ev[2]));
446        // Blocking copy: it waits for the kernel since both use stream 0.
447        CUDA_CHECK(cudaMemcpy(h_dst, g.d_dst, g.bytes,
448                              cudaMemcpyDeviceToHost));
449        CUDA_CHECK(cudaEventRecord(g.ev[3]));
450        CUDA_CHECK(cudaEventSynchronize(g.ev[3]));
```

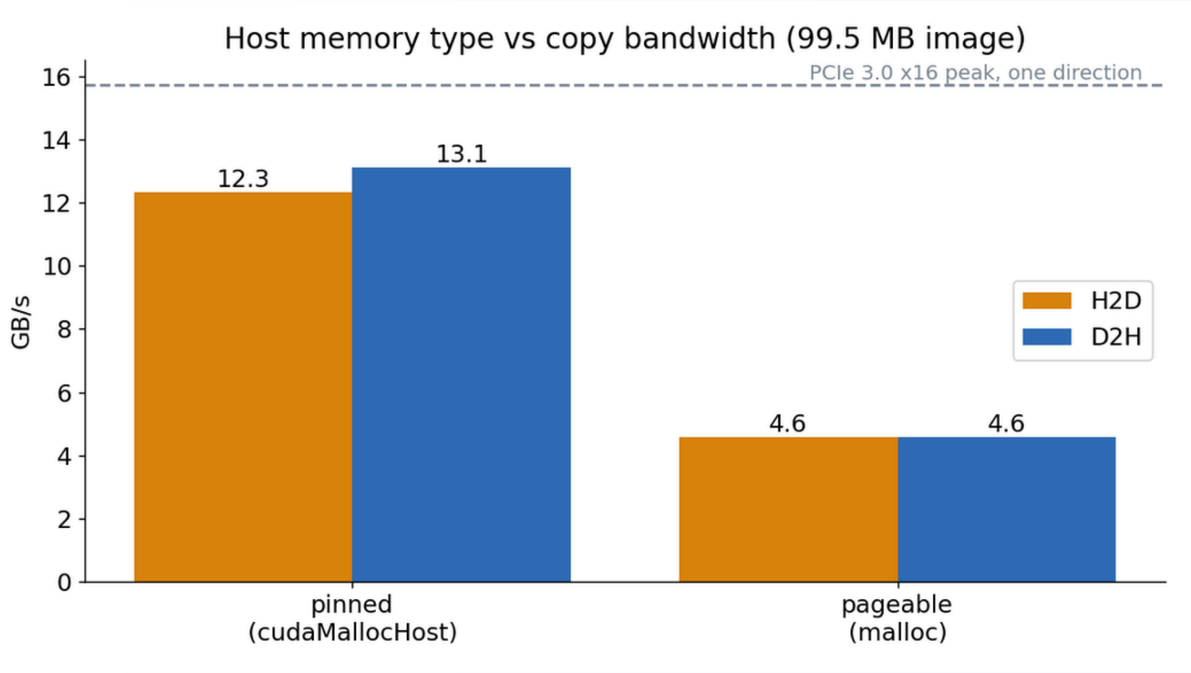
- Each stage timed on the GPU clock with events
- Warm-up run discarded, then mean of 10 runs
- --fmad=false: v1 to v5 match the CPU exactly



v6 texture differs in 257,964 bytes (0.26%), each by at most 1. Source: rotate_cuda.cu in our submission zip.

A5. Why PCIe is the narrow pipe

APPENDIX: extra material for Q&A, not part of the 7-minute talk



- PCIe 3.0 x16 (T4): 15.75 GB/s per direction
- Measured pinned copy: 12.3 GB/s
- T4 GDDR6: 320 GB/s
- About 25 FLOP per byte: rotation is memory bound
- PCIe is about 20x narrower, so copies are 91% of the time



Sources: PCI-SIG (2019); NVIDIA (n.d.-d). Roofline: T4 balance is about 25 FLOP per byte.

A6. GDS in practice

APPENDIX: extra material for Q&A, not part of the 7-minute talk

- Call order: cuFileDriverOpen, cuFileHandleRegister (file opened with O_DIRECT), cuFileBufRegister (GPU buffer), cuFileRead
- The CPU issues the calls; data DMAs NVMe to GPU with no host bounce buffer
- Needs nvidia-fs, O_DIRECT, aligned buffers, and a supported file system on local NVMe or NVMe-oF
- Without these, cuFile falls back to a host bounce buffer, which is what Colab would do



Sources: NVIDIA (n.d.-a, n.d.-b, n.d.-c).

A7. SIMD vs SIMT vs NPU

APPENDIX: extra material for Q&A, not part of the 7-minute talk

SIMD (CPU)	One instruction on a wide register, lanes in lockstep. Divergence is masked, which wastes lanes.
SIMT (GPU)	Warps of 32 threads run one instruction, each with its own index. Divergent branches serialise inside the warp.
NPU	Fixed matrix-multiply units (systolic array) for low-precision inference: high TOPS per watt, little flexibility. Rotation is a gather, so it maps poorly.



Sources: Lindholm et al. (2008); Jouppi et al. (2017).

A8. References, Task 1 (APA 7)

APPENDIX: extra material for Q&A, not part of the 7-minute talk

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Lindholm, E., Nickolls, J., Oberman, S., & Montrym, J. (2008). NVIDIA Tesla: A unified graphics and computing architecture. IEEE Micro, 28(2), 39-55. https://doi.org/10.1109/MM.2008.31

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Thompson, A., & Newburn, C. J. (2019, August 6). GPUDirect Storage: A direct path between storage and GPU memory. NVIDIA Technical Blog. https://developer.nvidia.com/blog/gpudirect-storage/



A9. References, Task 2 (APA 7), part 1

APPENDIX: extra material for Q&A, not part of the 7-minute talk

Ahuja, K., Diddee, H., Hada, R., Ochieng, M., Ramesh, K., Jain, P., Nambi, A., Ganu, T., Segal, S., Ahmed, M., Bali, K., & Sitaram, S. (2023). MEGA: Multilingual evaluation of generative AI. In Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (pp. 4232-4267). Association for Computational Linguistics. https://doi.org/10.18653/v1/2023.emnlp-main.258

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European Commission. (n.d.). Energy performance of data centres. https://energy.ec.europa.eu/topics/energy-efficiency/energy-efficiency-targets-directive-and-rules/energy-efficiency-directive/energy-performance-data-centres_en

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A9. References, Task 2 (APA 7), part 2

APPENDIX: extra material for Q&A, not part of the 7-minute talk

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